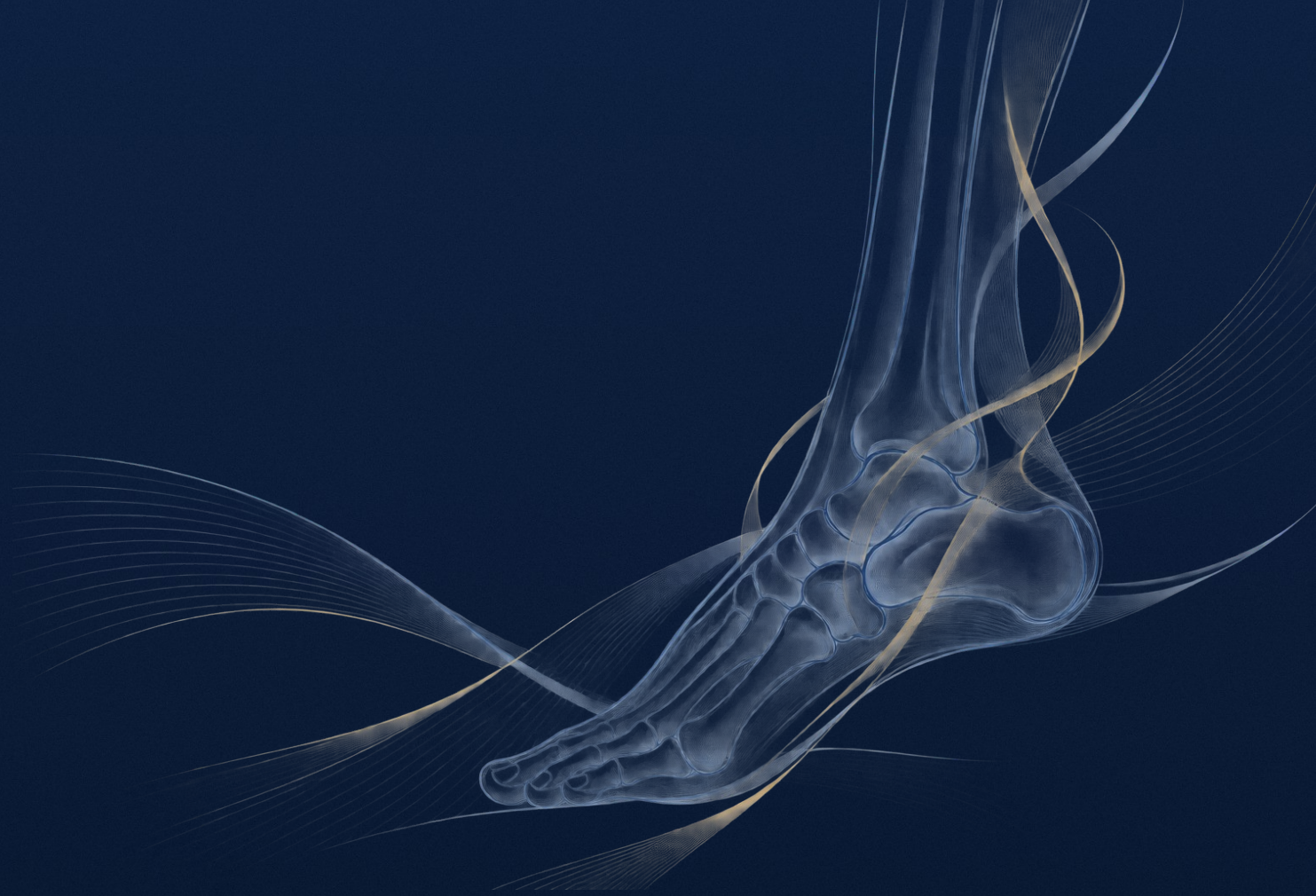
Effects of Foot Core and Trunk Core Training on Postural Control Strategies in Individuals with Chronic Ankle Instability

Jiyoon Lee

Department of Physical Education, Yonsei University

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Introduction



Chronic ankle instability and postural control

Up to 40% of first-time lateral ankle sprains progress to chronic ankle instability (CAI).

The hallmark deficit is postural control — most evident during dynamic tasks such as jump landing.

Deficits exist at both ends of the kinetic chain: the intrinsic foot muscles (distal) and the deep trunk stabilizers (proximal).

Individuals with CAI also show altered ankle, knee, hip and trunk kinematics during landing.

Two approaches, one open question

Foot core training (bottom-up) and trunk core training (top-down) both improve postural control in CAI.

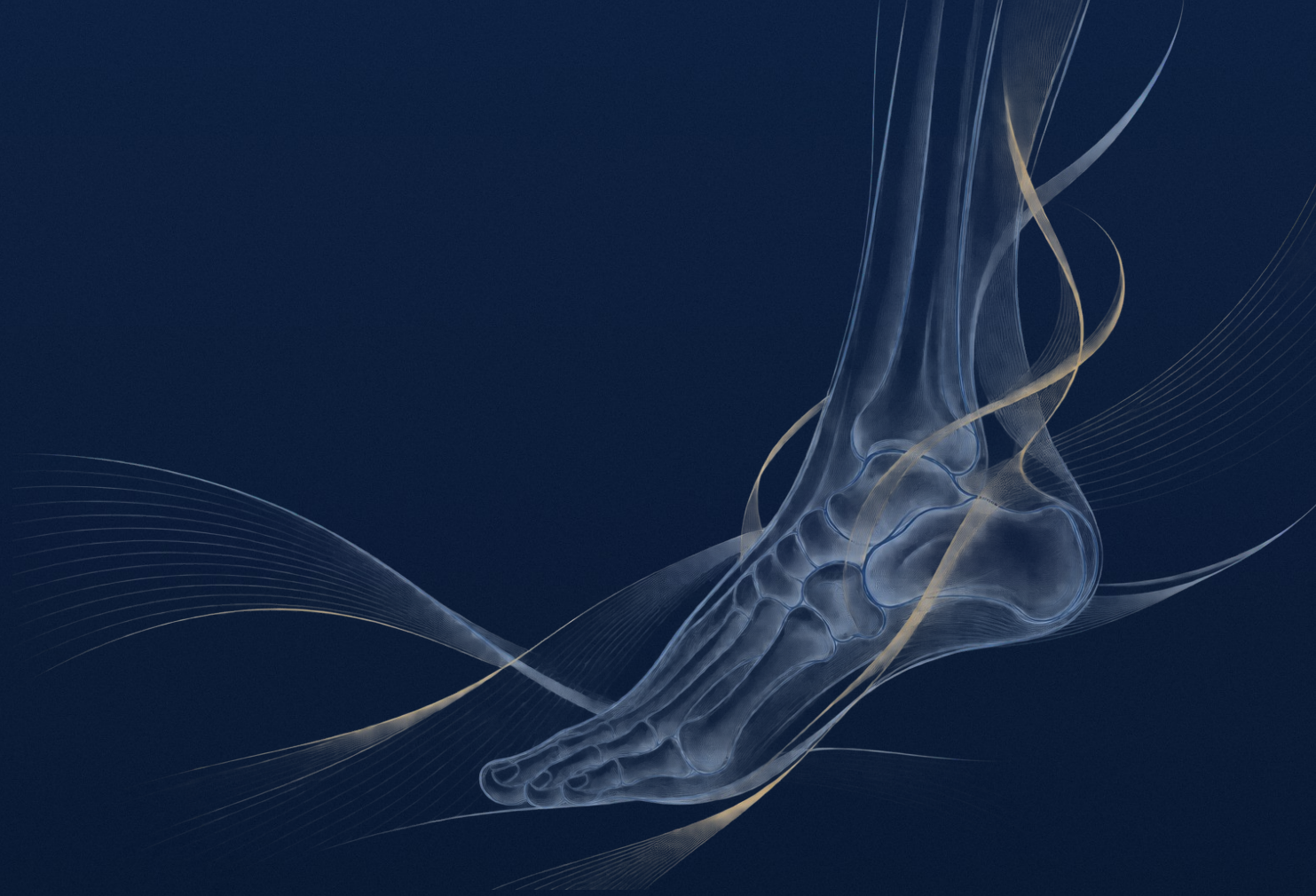
Whether either transfers to a dynamic landing task is unknown.

The two approaches have not been compared directly.

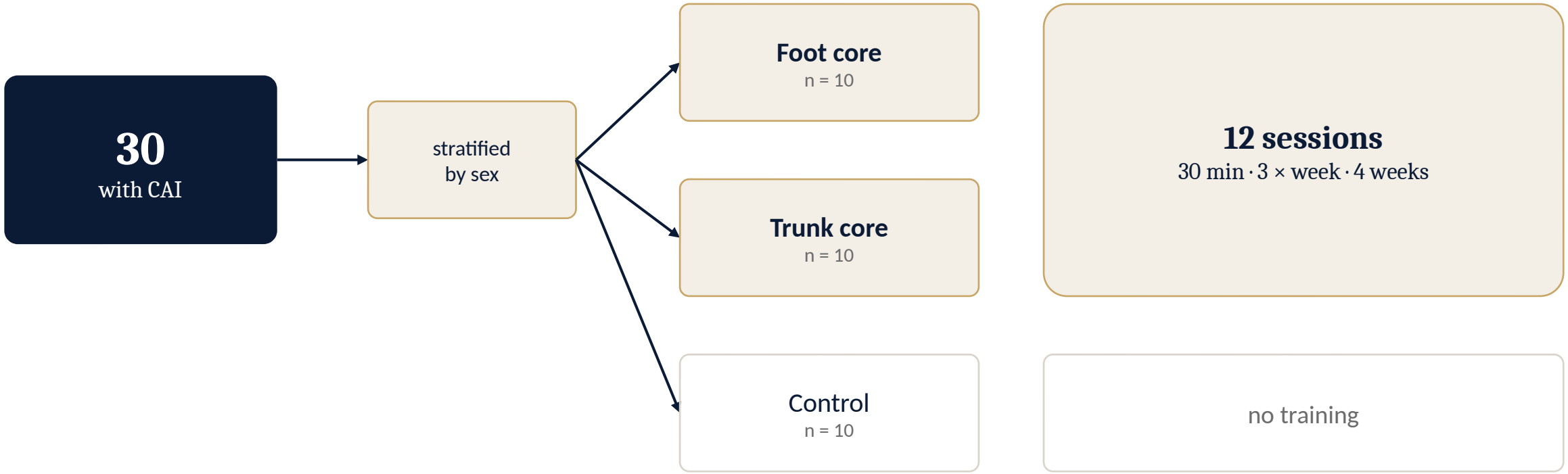
Purpose — to compare foot core training, trunk core training and a control condition on static and dynamic postural control, and on lower-limb and trunk kinematics, during single-leg landing in individuals with CAI.

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Methods



Study design



Randomized three-group pretest–posttest design · International Ankle Consortium criteria · IdFAI ≥ 11, FAAM-ADL < 90%, FAAM-Sport < 80%

Training programs

Foot core training (FCG)

[Exercise 1]

[Exercise 2]

[Progression]

12 sessions · 30 min · 3 × week · 4 weeks · supervised

Trunk core training (TCG)

[Exercise 1]

[Exercise 2]

[Progression]

12 sessions · 30 min · 3 × week · 4 weeks · supervised

Procedures and measures

Static postural control

Single-leg stance, involved limb, 10 s

Eyes open and eyes closed, 3 trials each

COP mean velocity (total, ML, AP)

95% ellipse area

Dynamic postural control and kinematics

Single-leg jump landing over a 30-cm hurdle from 40% body height

Dynamic postural stability index (DPSI), first 3 s after contact

3D kinematics of ankle, knee, hip and trunk (35 markers, 200 Hz)

Angle at initial contact and peak within 200 ms

Statistical analysis

Two-way mixed ANOVA — time (pre, post) × group (FCG, TCG, CG) — for discrete outcomes.

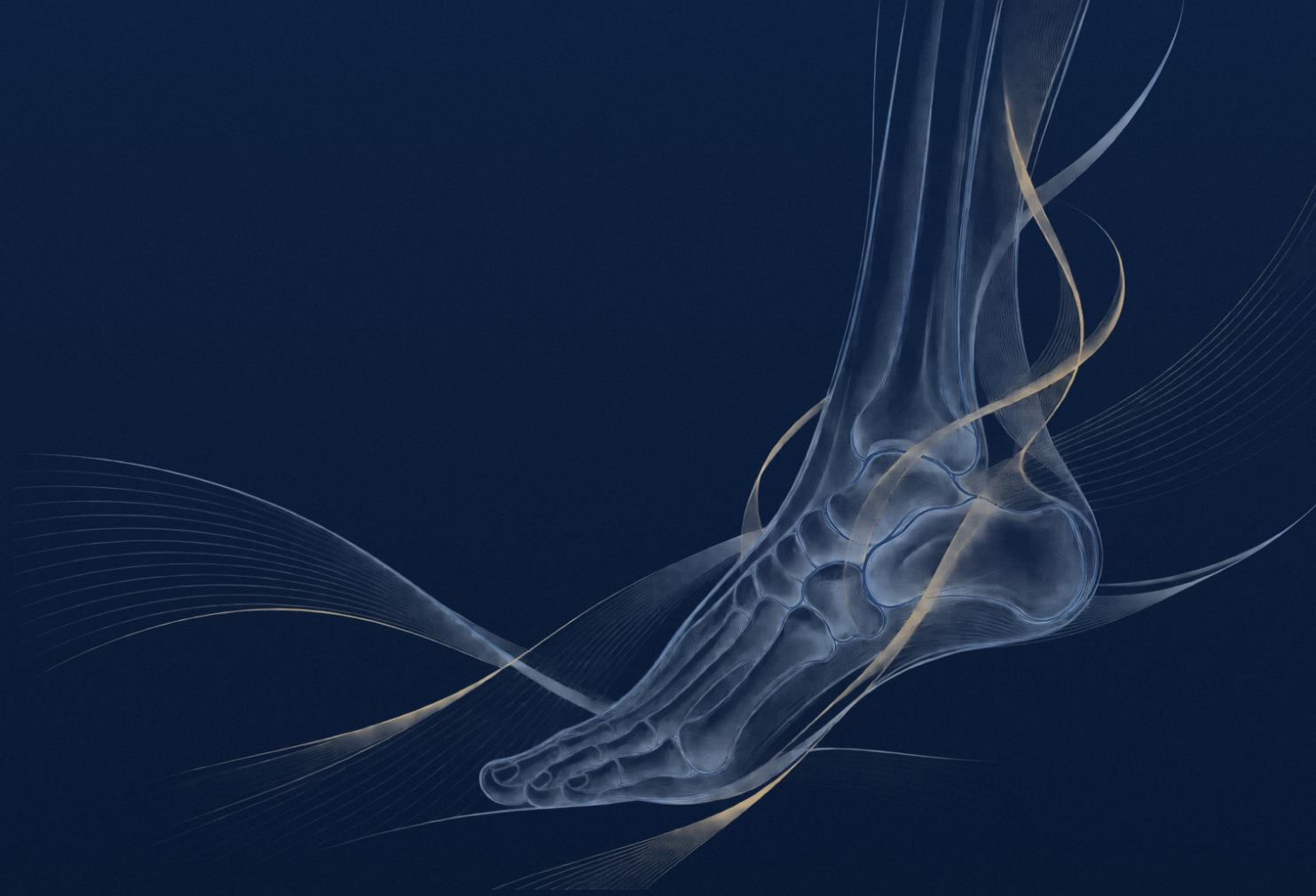
Significant interactions followed by one-way ANOVA on change scores with Bonferroni-adjusted pairwise comparisons.

Statistical parametric mapping (SPM) for continuous kinematics across the first 200 ms after initial contact; pairwise SPM t-tests at $p < 0.0167$.

Effect sizes as partial η^2; $\alpha = 0.05$.

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Results



Postural control

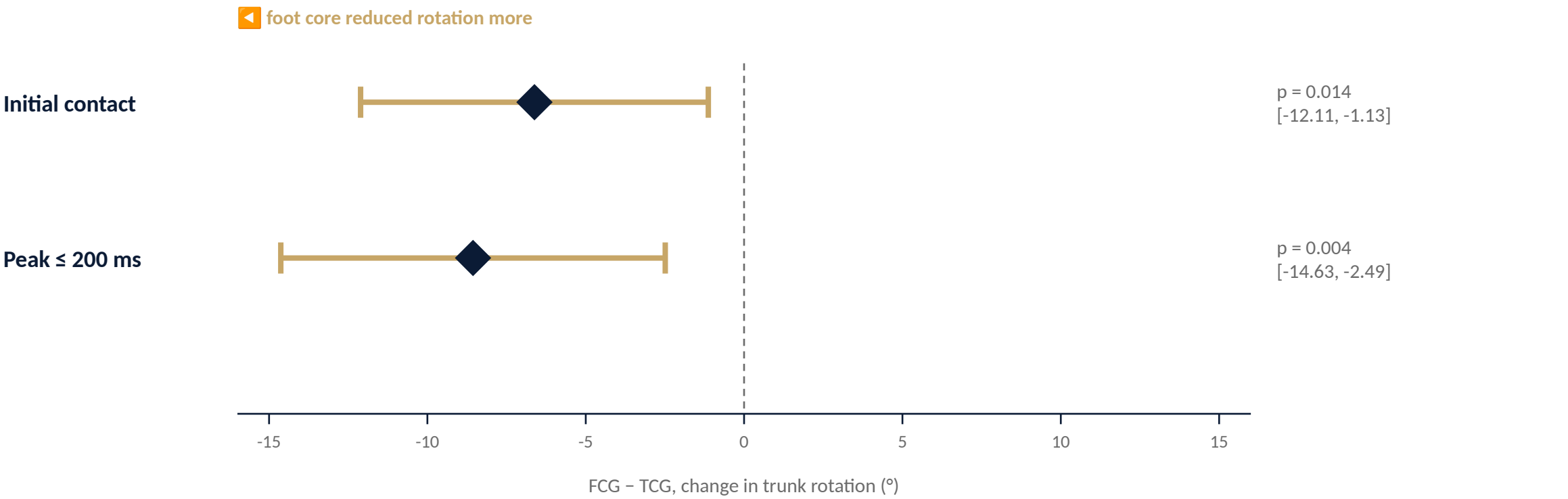
No baseline differences among groups in sex, age, height, weight, IdFAI or FAAM scores.

No significant time-by-group interaction for any static or dynamic postural control outcome (all $p > 0.05$).

Main effect of time only for 95% ellipse area, eyes closed ($p = 0.046$).

Table 2. Postural control outcomes, pre and post (to be inserted).

Trunk transverse-plane angle: time-by-group interaction



interaction IC $p = 0.016$, $\eta p^2 = 0.26$ · peak $p = 0.004$, $\eta p^2 = 0.34$ · FCG vs CG at peak $p = 0.053$ · TCG vs CG n.s. · ankle, knee, hip: no interaction

Statistical parametric mapping

Significant time-by-group interaction across the entire 0–200 ms interval (p = 0.010).

Greater reduction in trunk rotation in FCG than TCG from 25% to 100% of the interval (p = 0.010).

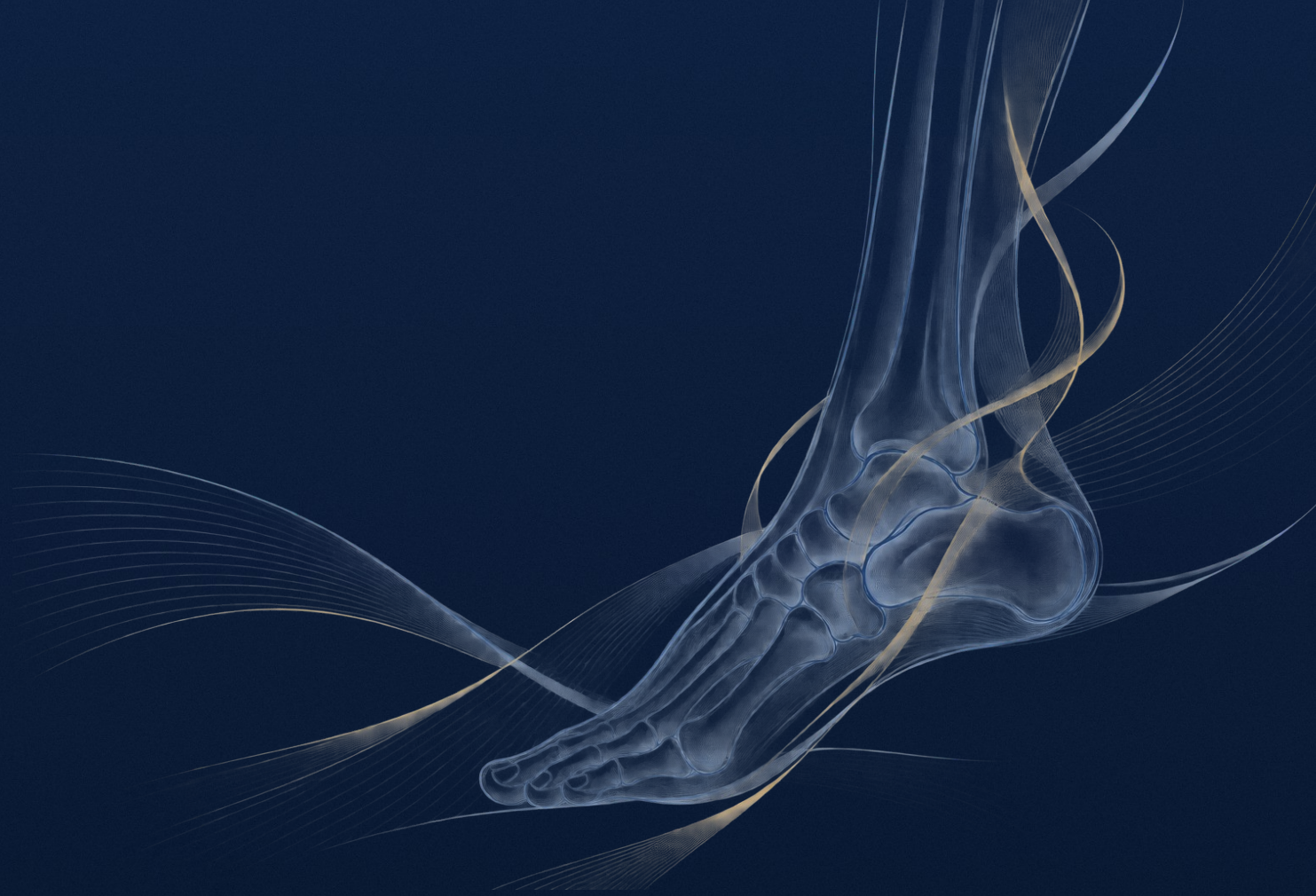
Greater reduction in FCG than CG from 80% to 100% (p = 0.015).

No difference between TCG and CG.

Figures 1–2. SPM interaction and pairwise change waveforms (to be inserted).

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Discussion



Main finding

Four weeks of foot core training reduced transverse-plane trunk rotation during single-leg landing — more than trunk core training over the same period.

Neither program changed static or dynamic postural control scores.

The change appeared in the movement strategy, not in the summary balance measures.

Interpretation

Why the foot changed the trunk

The foot is the first segment to contact the ground; the trunk is the last to settle.

Improved control at initial contact may reduce the correction required upstream.

Trunk stabilization in isolation may not transfer to a fast, single-leg landing.

Why balance scores did not change

Four weeks may be sufficient to shift a movement pattern but not a summary measure such as COP velocity.

Individuals with CAI may compensate to maintain stance while the underlying strategy differs.

Limitations

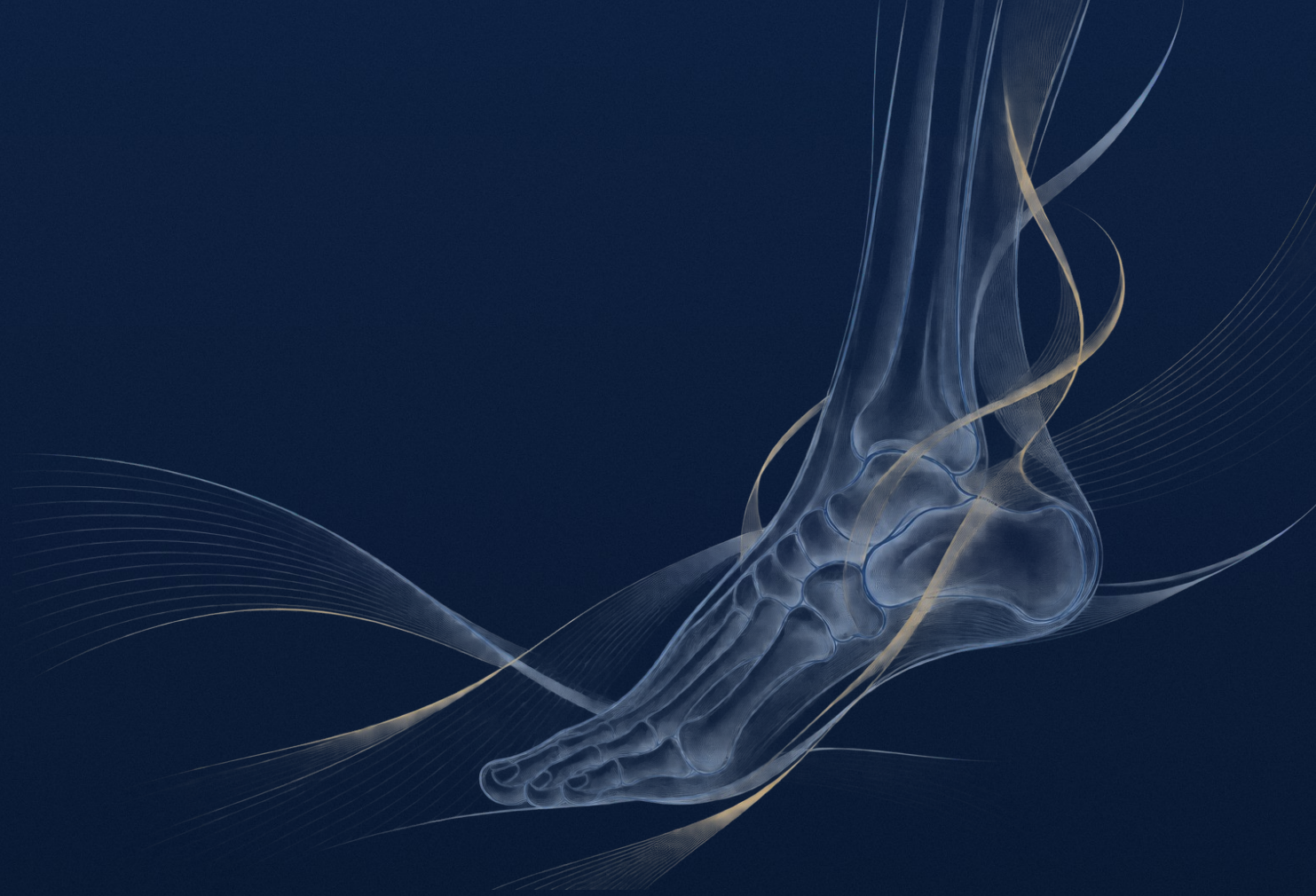
Sample size: 30 participants, 10 per group.

Intervention length: four weeks, no follow-up assessment.

Persistence of the kinematic change and its relation to giving-way episodes remain unknown.

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Conclusion



Conclusion and practical implications

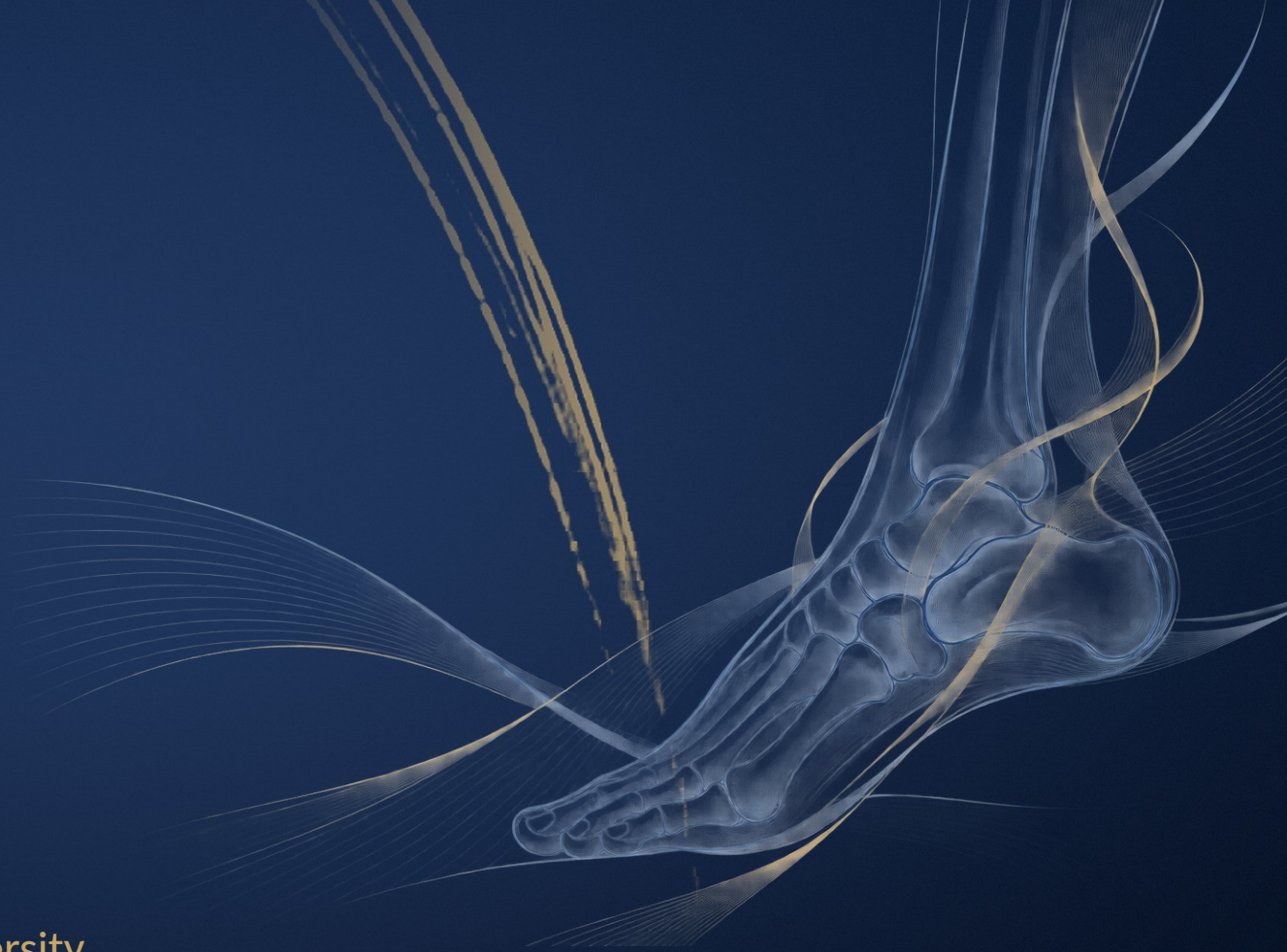
Four weeks of foot core training reduced trunk rotation during landing in individuals with CAI; trunk core training did not.

Foot core training may be a useful component for modifying altered landing patterns in CAI rehabilitation.

It is low-load, brief and supervised — twelve sessions of thirty minutes — and can be added alongside existing balance and strength work.

Thank You

Jiyoon Lee
[email]
Department of Physical Education, Yonsei University



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